11.5 Gravity in an Accelerating Train

WORKSHEET

The question

STEP II 2021, Question 10

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The train accelerates westwards (in the negative x direction) at a. For an observer in the train, what "fictitious" force appears to act on a mass m?

STAGE 1 No rest at the origin

Covers: Part (i)

Ask yourself If the bead stayed at the origin relative to the train, it would share the train's acceleration. What force would that need? At the origin the wire is horizontal: in which direction can its reaction act?

STAGE 2 An energy equation

Covers: Part (ii)

Pause and try Write Newton's second law in the train's frame, adding the force ma that comes from the frame's acceleration and including the normal reaction $\mathbf{N}$. Take the dot product with the velocity $(\dot{x}, \dot{y})$. Why does $\mathbf{N}$ drop out?

STAGE 3 How high?

Covers: Part (iii)

Ask yourself The constant is 0, since the bead starts at rest at the origin. At the highest point the bead is momentarily at rest. Use $ky = x^2$.

STAGE 4 The fastest point

Covers: Part (iv)

Ask yourself Write the speed squared as a function of x alone. Where is this quadratic greatest? Then find the gradient of the wire at that point.

What's really going on

Fictitious forces

Newton's laws hold in *inertial* frames, those that are not accelerating. In an accelerating frame they can be rescued by adding fictitious forces: $-m\mathbf{A}$ for a frame with acceleration $\mathbf{A}$, and, in a rotating frame, the centrifugal and Coriolis forces. They are called fictitious because no object exerts them; they are artefacts of the frame. But they are perfectly real to anyone inside it. The centrifugal force flattens the Earth at the poles, and the Coriolis force sets the direction in which hurricanes spin (anticlockwise in the northern hemisphere) and deflects long-range artillery shells.

The equivalence principle

A constant fictitious force $-m\mathbf{A}$ is proportional to mass, exactly like gravity. No experiment in a closed box can tell the difference between the box resting on a planet and the box accelerating in empty space: this is the *equivalence principle*. Einstein took it as the starting point of general relativity. Since light travelling across an accelerating box appears to bend, light must bend in a gravitational field too; and since clocks at the front and back of an accelerating rocket run at different rates, clocks at different heights in a gravitational field must too. Both predictions have been confirmed, and the second must be corrected for in every GPS satellite, whose clocks run faster than clocks on the ground by about 45 microseconds a day from this effect alone.

Einstein's happiest thought

In 1907, working at the patent office in Bern, Einstein was writing a review article on special relativity when, as he later recalled, "the happiest thought of my life" came to him: a person falling freely from a roof would not feel their own weight. Gravity could be "transformed away" by choosing a falling frame. It took him eight more

years, and a great deal of new mathematics (the differential geometry of curved spaces, developed by Gauss and Riemann), to turn the thought into the general theory of relativity of 1915.

Try this next

1. **A cannon on a train.** Try STEP I 2000, Question 9 (step.maths.org/questions/00-s1-q9): a projectile fired inside an accelerating carriage, where the acceleration acts like a tilt in gravity.

2. **A headwind.** Try STEP II 2006, Question 11 (step.maths.org/questions/06-s2-q11): a projectile against a constant wind, which can blow it back towards you before it lands.

3. **Coffee in a train.** Show that the surface of a cup of coffee in the accelerating train is a plane tilted at angle $\arctan(a/g)$ to the horizontal.

4. **Small oscillations.** Find the period of small oscillations of the bead about $x = \frac{ak}{2g}$. (*Hint:* near the new bottom the wire is approximately a circle of radius equal to its radius of curvature.)

Further reading

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), the chapter on non-inertial frames and fictitious forces.
- **Albert Einstein, *Relativity: The Special and the General Theory*** (various editions). Einstein's own popular account, including the accelerating box.
- **Pedro Ferreira, *The Perfect Theory*** (Little, Brown). A biography of general relativity.

The full guided solution is in Section 11.5:
https://beyondstep.keithmcnulty.org/ch11/accelerating-frame.html